

Repetition Control instructions.

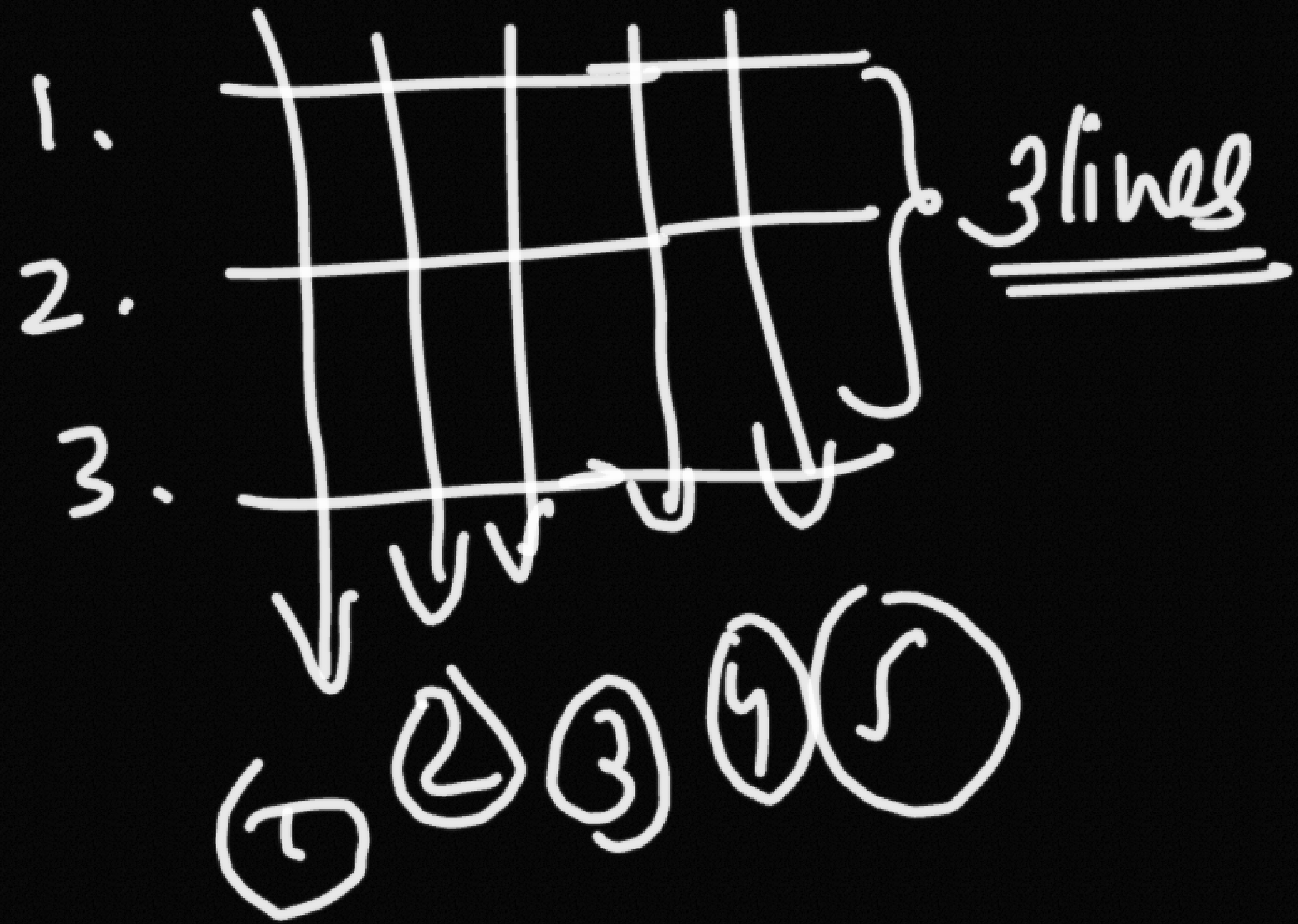
if we want to repeat statements
in the program, we use
repetition control instructions

There are two types
of repetition

Control instructions

1) while loop

2) for loop

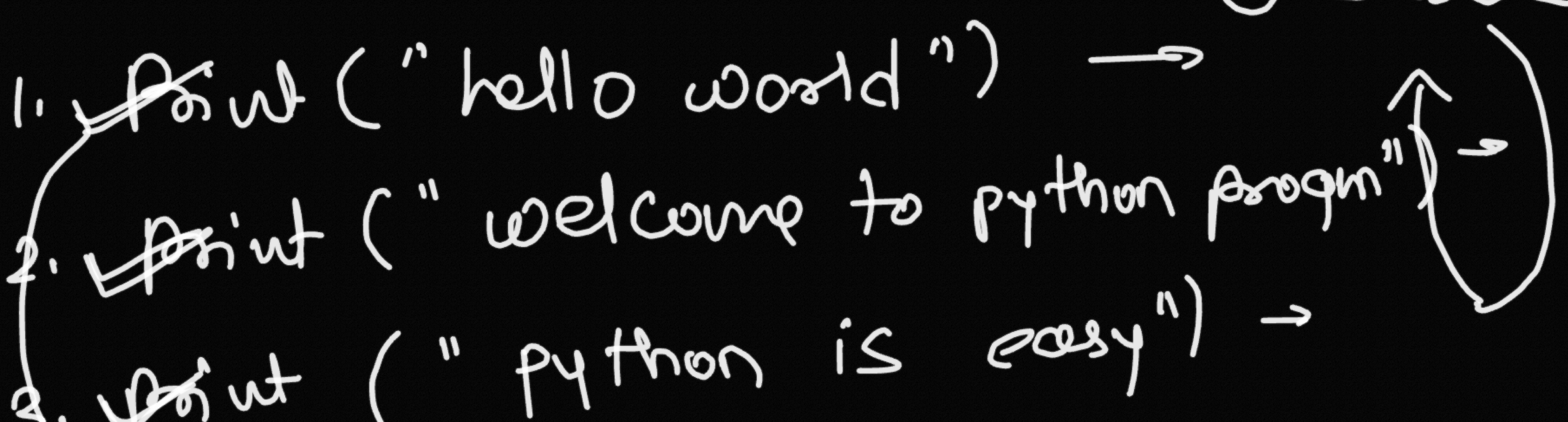


① ② ③ ④ ⑤

1. ~~print~~ ("hello world") →

2. ~~print~~ ("welcome to python prog") →

3. ~~print~~ ("python is easy") →



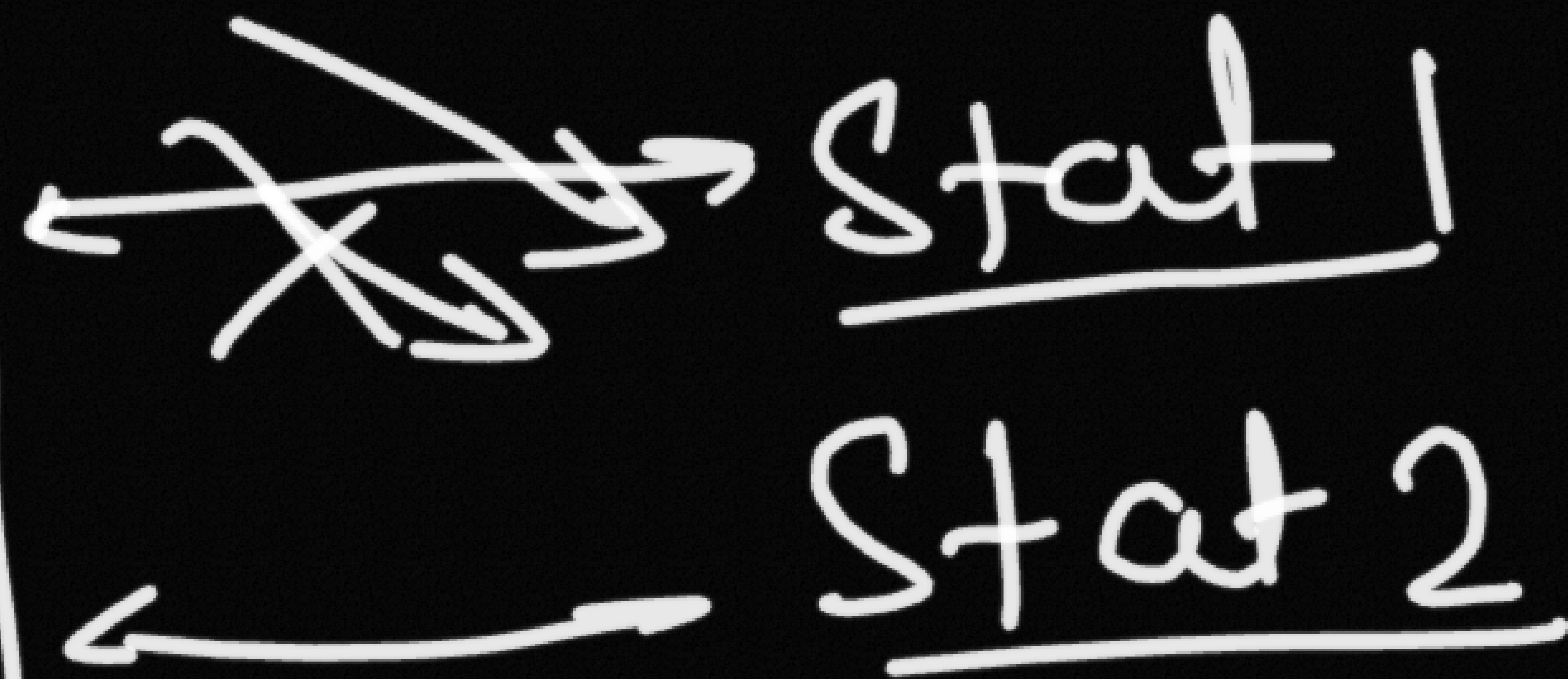
h w
w t p y
p i e
h w
w t p y
p i e

if while loop -

Syntax:

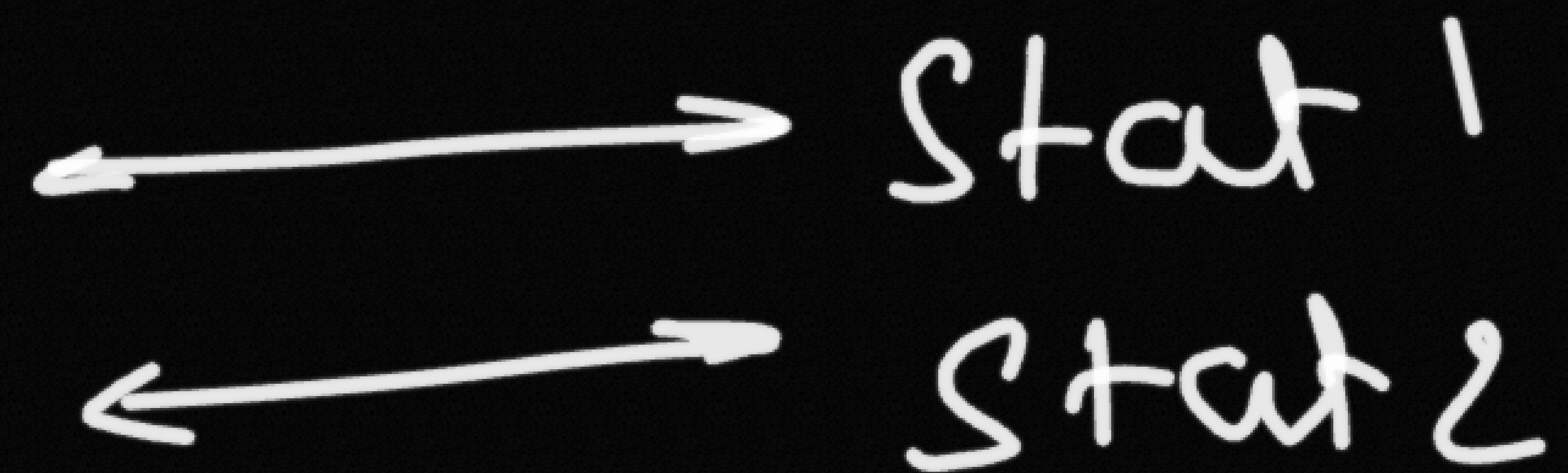
①

while <Condition>:



②

while <Condition>:



else:

Stat 1
Stat 2

- 1) until condition is true, statements inside while block executes repeatedly
- 2) else block is optional but if present, when while condition is false, else block will execute.

3) if while block terminated abruptly using break statement, else block will not execute.

count = 0 6 <= 5
while count <= 5 :

print(count)

count = count + 1

Print ("Program executed")

O/P -

~~0~~
~~1~~
~~2~~
~~3~~
~~4~~
~~5~~

It 1
~~0 <= 5~~
= 0 + 1
count = 1

It 2
~~1 <= 5~~
= 1 + 1
count = 2

It 3
~~2 <= 5~~
= 2 + 1
count = 3

3 + 4

~~1 + 5~~ ~~4 <= 5~~
= 4 + 1
5
~~3 <= 5~~
= 5 + 1
~~1 + 4~~
~~3 <= 5~~

for loop :-

1) for loop is used to iterate
over sequence

Syntax:

for var in seq:

 stat 1

 stat 2

name = "ganesh"

for var in name:

 print(var)

o/p- g
 n
 e
 s
 h

String is sequence or not ?

name = "ganesh"
 0 1 2 3 4 5
 ↑ ↑ ↑ ↑ ↑ ↑
print(name) → ganesh

String → Sequence ✓

① List $\rightarrow$ Sequence

list1 = [10, 20, 30, 40, 50]

② Tuple $\rightarrow$ Sequence

tuple1 = (10, 20, 30, 40, 50)

③ set — does not maintain seq.

set = {10, 20, 30, 40, 50}

④ dictionary — sequence.

dict1 = {"name": "ganesh",
 "age": 25}

range() — Built in function.

This is the built in function of python & used to generate sequence of numbers.

eg. 1-10, 50-100, 20-40

2) we commonly use it in for loop.

Syntax: $\begin{matrix} 1 & 10 \\ \text{range}(\text{start}, \text{stop}^{+1}, \text{step}) \end{matrix}$

• `range(1, 11, 1)`

o/p- 1 2 3 4 5 6 7 8 9 10

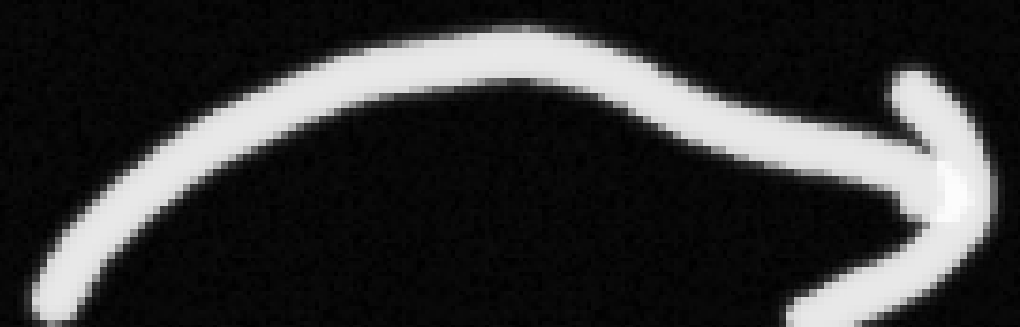
• `range(50, 101, 1)`

o/p- 50, 51, 52 - - - 100

range (20, 41, 1)

o/p 20, 21, 23, --- 40

range (2, 11, 2)

o/p 2, , 4, 6, 8, 10

for i in range(1, 11, 1):
 print(i)

O/P -

1	8
2	9
3	10
4	
5	
6	
7	
8	
9	
10	

Break Statement -

1. we can use break Statement in while loop & for loop.
2. when it executes break Statement it terminates loop immediately, & executes the next statement in the program.

for i in range(1, n+1):

print(i)

o/p -
1
2
3
⋮
10


```
for i in range(1, 11, 1):  
    if i == 5:  
        break  
    print(i)
```

o/p -

Continue -

1) we can use continue in for & while loop.

2) Continue statement is used to skip current iteration & keep executing next iteration.

